

AGYEI MICHAEL JUNIOR

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📍 Accra, Ghana

🔄 agyeimchjunior-create

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Profile

Recent Materials Science and Engineering graduate with a solid academic foundation and hands-on experience in materials characterization, testing, and processing. I have completed my national service and I am currently seeking employment or internship opportunities. I am Skilled in translating UI/UX designs into clean, reusable components and optimizing performance for modern browsers. Strong understanding of component-based architecture, routing, and state management, with a passion for creating visually appealing and intuitive user interfaces. Comfortable collaborating with backend developers and working in Git-based workflows.

Education

2021 – 2024 Accra, Ghana	Bsc. Material Science in Engineering <i>University Of Ghana</i> Bsc in Material Science in Engineering with, Second Class honours(upper division) Relevant courses: C programming , Matlab, Chemistry, Physics, Material selection, Polymer processing, Quality control, Phase equilibria, Applied Electricity
2017 – 2020 Sunyani, Ghana	General Science <i>St. James Seminary and Senior High School</i> Learnt some important courses like biology, physics, chemistry, ICT, mathematics and social studies.

Professional Experience

10/2024 – 10/2025 Accra, Ghana	Teaching Assistant <i>University of Ghana</i> • Organizing tutorials sessions for students and helping them with their work, and also providing individualized support and guidance • Assisting my lecturer with his ongoing projects and lesson planning
02/2023 – 07/2023	Lab Technician Intern <i>University of Ghana Engineering Mechanical lab</i> • I was responsible for the proper sterilization of all glassware and tools, safe disposal of chemical waste, and strict adherence to Safety Data Sheet (SDS) guidelines • Keeping perfect records and observing measurements and equipment by complying with Good Laboratory Practice requirements.
10/2023 – 01/2024 Dormaa Ahenkro, Bono Region	Teaching Intern <i>Delany Academy</i> • Delivering lessons, guide students through learning activities and evaluate their progress

- Planning and preparing lessons, including setting goals, objectives and learning outcomes.

10/2022 – 01/2023

Sales Person

Dormaa Ahenkro, Bono
Region

Oduro Pharmacy

- Assist customers with their queries, provide product information, and help them find good products.
- Operate the cash register and handle customer payments

Research Interest

- Software development
- Polymers
- Sustainable waste management practises
- Web development and Animations

Skills

Frontend Development — : HTML5, CSS3, JavaScript (ES6+), TypeScript, React.js, React , GSAP, |
Styling: Tailwind CSS, Bootstrap, Responsive Design | **Tools/Workflow:** Git/GitHub

Backend Development — Node.js, Express.js, Mongo DB, Supabase, JWT Authentication, API Security & Middleware

Graphic Designing : Adobe Photoshop

Proficient in Microsoft tools — Excel, Microsoft Word, Power point

UI/ UX Designing : Figma — Figma

Research and Development — conducted research on plastics(LDPE/HDPE), waste tires and polymers and how they can be managed sustainably to promote a greener environment.

Game developer — Experienced in using **Scratch (block-based coding)** to teach foundational programming concepts and logic to children. Developed engaging, educational games and interactive stories tailored for various age groups.

Soft Skills — | I maintain exceptional standards of quality and detail in all my work. | I am completely reliable, always on time, and manage my schedule efficiently. | You can trust me for complete honesty and discretion with all property. | I don't wait for direction; I see problems and take the correct action without hesitation. | I quickly master new processes and procedures, and I am committed to learning rapidly.

Leadership Experience

01/2024 – 09/2024

Committee Head of Welfare

University of Ghana(Engineering school)

- Managing the welfare committee's budget, resources, and materials required to run their programs.
- Supervising committee members to check up on students mental health and how they are catching up with their academic journey.

01/2023 – 09/2024

Floor rep

University of Ghana(kwapong Hall)

Organising floor meetings and reporting the needs of floor members to higher offices for solutions

Health Prefect

St. James Seminary

- Helping students with first aid
- Promoting hygiene and showcasing leadership and mentorship

Projects

01/2024 – 2024	Conversion of Waste tires into biofuel, char and gas using pyrolysis <i>University of Ghana</i> • waste tires were collected in different sources in Accra around Madina and legon campus • These tires were cleaned and cut into smaller sizes before they were shredded • The shredded tires undergoes a scientific process called pyrolysis where they are been broken down at elevated temperatures. • The liquid fraction(biofuel) is been collected and also the solid fraction(char) is also been collected and the gas is also collected in tube
05/2024 – 10/2024	Plastic Waste Valorization Facility For Fuel Production <i>Osu Alata, Landing Beach</i> • Plastics wastes were collected from and around the beach • They were washed and dried before undergoing pyrolysis • At higher temperatures these plastics waste are converted into biofuel, gas and char • The fuel collected was given to the fisherman at the shore to power their canoe engine for fishing
02/2023 – 05/2023	Synthesizing Nano Particles for Energy Storage <i>University of Ghana Mechanical Lab</i> This project focussed on synthesizing nanoparticles, specially vanadium pentaoxide(V₂O₅) and Nickel Oxide(NiO), utilizing hydrotherma and precipitation methods, respectively. Characterization of the synthesized nanoparticles was conducted through techniques such as X-ray diffraction, scanning electron microscopy for energy storage devices.
09/2025 – 01/2026	Full-Stack Data Platform • Built user registration and email verification system • Integrated Supabase authentication and PostgreSQL database • Implemented role-based access control • Connected frontend (React) to secure backend APIs
01/2026 – 02/2026	Custom Admin Dashboard E-Commerce Backend API • Designed and developed RESTful APIs using Node.js and Express • Implemented JWT-based authentication and protected routes • Integrated MongoDB for product, user, and order management • Deployed backend service on Render

References

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